

Executive Summary — WCDN Aegis Automation Protocol

The purpose of this project was to simulate a real-world cloud security incident involving a compromised EC2 instance attempting to communicate with a known malicious entity. The operation focused on leveraging AWS native security tools—GuardDuty, Security Hub, CloudTrail, VPC Flow Logs, Lambda, and SNS—to detect, analyze, and automatically remediate threats in a controlled cloud environment.

The mission began by deploying two EC2 instances representing the Hydra-Infiltrator (victim) and the Loki-Variant (malicious actor), along with correctly configured VPC networking, security groups, and key pairs. After enabling GuardDuty and enriching it with a custom threat list uploaded to S3, the environment was prepared to detect suspicious activity and correlate threats through AWS Security Hub, which provided centralized visibility and normalization of findings.

To simulate the attack, outbound communication was initiated from the Hydra-Infiltrator to the malicious Loki-Variant. This activity triggered GuardDuty alerts, which were forwarded to Security Hub for analysis. Logging and investigation visibility were enhanced using CloudTrail to track API activity, and VPC Flow Logs to verify network traffic patterns, detect connections, and observe potential command-and-control behavior.

A key part of the project was designing and deploying an automated incident response pipeline. This included:

- SNS: instant email alerts for GuardDuty findings
- Lambda: automatic stopping of compromised EC2 instances based on GuardDuty events
- EventBridge: the trigger mechanism connecting GuardDuty alerts to the Lambda containment function

This automation created a seamless workflow: Threat Detected → Alert Sent → EC2 Instance Automatically Stopped.

The project concluded with a complete security cleanup to eliminate resources and prevent ongoing AWS charges, including terminating EC2 instances, deleting Elastic IPs, removing the S3 threat list bucket, decommissioning Lambda and IAM roles, and disabling GuardDuty, Security Hub, CloudTrail, flow logs, alarms, and associated IAM permissions.

Overall, this project demonstrated end-to-end cloud security capability, including secure architecture design, threat detection, centralized monitoring, network and API forensics, and fully automated incident response. It reflects a strong understanding of cloud defense strategy, AWS security tools, automation mechanisms, and real-world response workflows.